

Lecture 21: Gradient fields and potential functions

Testing whether $\vec{F} = \langle M, N \rangle$ is a gradient field

If $\vec{F} = \nabla f$: $M = f_x$ then $f_{xy} = f_{yx} \Leftrightarrow M_y = N_x$

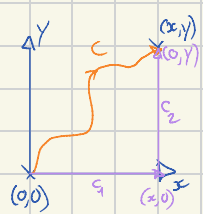
Conversely: if $\vec{F} = \langle M, N \rangle$ is defined, differentiable everywhere in the plane and $M_y = N_x$ then $\vec{F}$ is a gradient field.

Example: $\vec{F} = \underbrace{-y^1 + x^1}_M \hat{i} + \underbrace{0}_N \hat{j}$ $\frac{\partial M}{\partial y} = -1 \neq 1 = \frac{\partial N}{\partial x}$ Therefore $\vec{F}$ is not a gradient field

* $\vec{F} = \underbrace{(4x^2 + axy)}_M \hat{i} + \underbrace{(3y^2 + 4xz^2)}_N \hat{j}$ For which value of "a" is this a gradient field? $a=8$ $M_y = ax$ $N_x = 8x \Leftrightarrow a=8$

Finding the potential? (only if $M_y = N_x$)

① Computing line integrals



$$\int_C \vec{F} \cdot d\vec{x} = f(x_1, y_1) - f(0,0) \quad f(x_1, y_1) = \int_C (\vec{F} \cdot d\vec{x}) + \underbrace{f(0,0)}_{\text{constant}}$$

$$\int_C \vec{F} \cdot d\vec{x} = \int_C (4x^2 + 8xy) dx + (3y^2 + 4xz^2) dy$$

On C_1 : $0 \leq x \leq x_1$ $y=0, dy=0$ then: $\int_{C_1} \vec{F} \cdot d\vec{x} = \int_0^{x_1} 4x^2 dx = \frac{4}{3} x_1^3$

On C_2 : y from 0 to y_1 $x=x_1, dx=0$ then $\int_{C_2} \vec{F} \cdot d\vec{x} = \int_0^{y_1} (3y^2 + 4x_1^2 y) dy = \left[y^3 + 4x_1^2 y \right]_0^{y_1} = y_1^3 + 4x_1^2 y_1$

So $f(x_1, y_1) = \frac{4}{3} x_1^3 + y_1^3 + 4x_1^2 y_1 + C$

② Antiderivatives: We want to solve: $\begin{cases} f_x = 4x^2 + 8xy & \textcircled{A} \\ f_y = 3y^2 + 4x^2 & \textcircled{B} \end{cases}$

$\textcircled{A} \Rightarrow f = \frac{4}{3} x^3 + 4x^2 y + g(y)$

$f_y = 4x^2 + g'(y)$ Make it match with $\textcircled{B}$

$4x^2 + g'(y) = 3y^2 + 4x^2 \Leftrightarrow g'(y) = 3y^2 \Leftrightarrow g(y) = y^3 + C$ So: $f = \frac{4}{3} x^3 + 4x^2 y + y^3 + C$

$\vec{F} = \langle M, N \rangle$ is a gradient field in a region of the plane

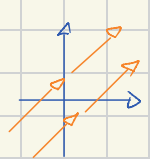
Conservative:

$\oint_C \vec{F} \cdot d\vec{x} = 0$ for closed curve $N_x = M_y$ at any point

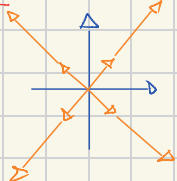
Definition: $\text{curl } \vec{F} = N_x - M_y$ so $\text{curl } \vec{F} = 0$ (test for conservativeness: $\text{curl } \vec{F} = 0$)

The sign indicates the direction of rotation (counterclockwise/clockwise)

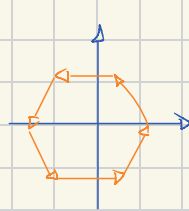
* For a velocity field: curl measures rotation component of motion



$\vec{F} = \langle a, b \rangle$, $\text{curl } \vec{F} = 0$



$\vec{F} = \langle x, y \rangle$ $\text{curl } \vec{F} = \frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x) = 0$



$\vec{F} = \langle -y, x \rangle$ $\text{curl } \vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) = 2$

curl measures (2 times) angular velocity of rotation component of velocity field

curl of a force field measures torque exerted on a test object in the field.

$\frac{\text{torque}}{\text{moment of inertia}} = \frac{d}{dt} (\text{angular velocity})$

$\frac{\text{force}}{\text{mass}} = \frac{d}{dt} (\text{velocity})$